

Equity Valuation

A Comprehensive Interdisciplinary Study Guide

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Contents

1	Introduction	4
2	The Fundamental Valuation Identity	4
3	Enterprise Value and Equity Value	5
4	Cost of Equity and the Required Return	5
4.1	Capital Asset Pricing Model	5
4.2	Cost of Debt	5
4.3	WACC	5
4.4	Example	6
5	Dividend Discount Models	6
5.1	Gordon Growth Model	6
5.2	Example	6
5.3	Multi-Stage Dividend Model	6
5.4	H-Model	7
5.5	Three-Stage Dividend Model	7
6	Free Cash Flow to Equity	7
6.1	Example	7
7	Free Cash Flow to Firm	8
7.1	Example	8
8	Economic Profit and EVA	8
9	Residual Income Valuation	9
9.1	Example	9
10	Abnormal Earnings Growth	9
11	Price Multiples	9
11.1	Price-to-Earnings	10
11.2	Forward P/E	10
11.3	PEG Ratio	10
11.4	Price-to-Book	10
11.5	Price-to-Sales	10
12	Enterprise Value Multiples	11
12.1	EV/EBITDA	11
12.2	EV/EBIT	11
12.3	EV/Sales	11
12.4	EV/FCFF	11

13 Comparable Company Analysis	11
14 Precedent Transactions	12
15 Sum-of-the-Parts Valuation	12
16 Asset-Based Valuation	12
16.1 Adjusted Net Asset Value	13
16.2 Liquidation Value	13
17 Replacement-Cost Valuation	13
18 Tobin's Q	13
19 Venture Capital Method	14
20 Probability-Weighted Valuation	14
21 Decision-Tree Valuation	14
22 Real Options Valuation	15
23 Abandonment Option	15
24 Adjusted Present Value	16
25 Leveraged Buyout Valuation	16
25.1 Example	16
26 Market Capitalization from Share Repurchases	17
27 Implied Valuation and Reverse DCF	17
28 Dividend Yield and Earnings Yield	18
29 Residual Asset and Liquidation Valuation	18
30 Equity as a Call Option on the Firm	18
31 Black–Scholes–Merton Structural Valuation	19
32 Conglomerate Discount and Holding-Company Valuation	19
33 Private Company Valuation	19
34 Financial Institution Valuation	20
35 Insurance Company Valuation	20
36 Commodity and Resource Companies	20
37 Technology Company Valuation	20
38 Platform and Network Valuation	21
39 Human Capital and Intangible Assets	21
40 Growth Decomposition	21
41 Value Creation and Competitive Advantage	22
42 Terminal Value	22

43 Sensitivity Analysis	22
44 Monte Carlo Valuation	23
45 Bayesian Valuation	23
46 Factor-Based Relative Valuation	23
47 Regression-Based Valuation	24
48 Market-Implied Expected Return	24
49 Cross-Asset Relative Valuation	24
50 Behavioral Finance Perspective	25
51 Game-Theoretic Perspective	25
52 Macroeconomic Perspective	26
53 Accounting Perspective	26
54 Information-Theoretic Perspective	27
55 Uncertainty and Knightian Uncertainty	27
56 Real-Options and Strategic Management	27
57 A Unified Valuation Framework	27
58 Worked Integrated Example	28
58.1 Comparable Valuation	29
58.2 Residual Income Valuation	30
59 Why Valuation Methods Disagree	30
60 Valuation Triangulation	31
61 Common Valuation Errors	31
61.1 Terminal Growth Greater Than Discount Rate	31
61.2 Mixing FCFF and Cost of Equity	31
61.3 Double Counting Cash	31
61.4 Ignoring Dilution	31
61.5 Confusing Growth With Value Creation	31
61.6 Mechanical Use of Multiples	32
62 A Practical Valuation Workflow	32
63 Interdisciplinary Synthesis	32
64 The Deep Structure of Equity Valuation	33
64.1 Question 1: What does the shareholder own?	33
64.2 Question 2: What economic benefits will the claim generate?	33
64.3 Question 3: When will those benefits occur?	33
64.4 Question 4: What risks and alternatives determine their value?	33
65 Master Valuation Map	33
66 Final Formula Sheet	33
67 Conclusion	35

1 Introduction

Equity valuation is the process of estimating the economic value of an ownership claim on a business. The central difficulty is that equity is not usually a directly observable object with a unique intrinsic price. Instead, its value must be inferred from future cash flows, accounting profits, assets, market prices, transactions, strategic opportunities, and the risks associated with alternative states of the world.

A useful organizing principle is

$$\boxed{\text{Equity Value} = \text{Value of Expected Economic Benefits}}$$

subject to the risk, timing, financing structure, and strategic opportunities associated with those benefits.

There are therefore several logically distinct valuation traditions:

1. discounted cash-flow valuation;
2. dividend valuation;
3. relative valuation;
4. residual-income and economic-profit valuation;
5. asset-based valuation;
6. transaction-based valuation;
7. option and contingent-claim valuation;
8. probability-weighted and scenario valuation;
9. venture-capital and entrepreneurial valuation;
10. leveraged-buyout valuation;
11. strategic and real-option valuation;
12. reverse valuation and implied-expectation analysis.

No single method dominates every circumstance.

2 The Fundamental Valuation Identity

For any financial claim whose future payoff is represented by X_t , the most general valuation expression is

$$V_0 = E_0 \left[\sum_{t=1}^T \frac{X_t}{(1+r_t)^t} \right].$$

In a more general stochastic setting,

$$V_0 = \sum_s \Pr(s) \frac{X(s)}{D(s)},$$

where s indexes possible states of the world and $D(s)$ is the appropriate state-price denominator. Under risk-neutral valuation,

$$V_0 = E^Q \left[\frac{X_T}{R_f^T} \right].$$

The different valuation methods can therefore be understood as different ways of specifying:

1. what constitutes the payoff;
2. how the payoff evolves;

3. how uncertainty is represented;
4. how risk is priced;
5. how terminal value is represented.

3 Enterprise Value and Equity Value

A central distinction is

$$EV = \text{Value of the operating enterprise}$$

and

$$\text{Equity Value} = EV - Debt + Cash$$

in the simplest formulation.

More generally,

$$\frac{\text{Equity Value}}{\text{Value}} = EV - Debt - \frac{\text{Preferred Equity}}{\text{Equity}} - \frac{\text{Noncontrolling Interests}}{\text{Interests}} + \frac{\text{Excess Cash}}{\text{Cash}} + \frac{\text{Nonoperating Assets}}{\text{Assets}}.$$

Per-share value is

$$P_0 = \frac{\text{Equity Value}}{\text{Diluted Shares Outstanding}}.$$

This distinction is crucial because operating cash flows generally belong to both debt and equity holders, whereas free cash flow to equity belongs only to common shareholders.

4 Cost of Equity and the Required Return

4.1 Capital Asset Pricing Model

The classical CAPM is

$$r_e = r_f + \beta_e(E[R_m] - r_f).$$

For an international company one may augment the model with a country risk premium:

$$r_e = r_f + \beta_e ERP + CRP.$$

Here,

- r_f is the risk-free rate;
- β_e is equity beta;
- ERP is the market equity-risk premium;
- CRP is a country-risk premium.

4.2 Cost of Debt

The after-tax cost of debt is

$$r_d^{AT} = r_d(1 - T).$$

4.3 WACC

For a conventional enterprise,

$$WACC = \frac{E}{D + E} r_e + \frac{D}{D + E} r_d(1 - T).$$

4.4 Example

Suppose

$$E = 600, \quad D = 400,$$

with

$$r_e = 10\%, \quad r_d = 6\%, \quad T = 25\%.$$

Then

$$WACC = 0.60(0.10) + 0.40(0.06)(1 - 0.25)$$

and therefore

$$\boxed{WACC = 7.8\%}.$$

5 Dividend Discount Models

The dividend discount model values equity directly:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_e)^t}.$$

5.1 Gordon Growth Model

If dividends grow forever at constant rate g ,

$$\boxed{P_0 = \frac{D_1}{r_e - g}}$$

provided

$$r_e > g.$$

5.2 Example

Suppose

$$D_0 = 4, \quad g = 5\%, \quad r_e = 11\%.$$

Then

$$D_1 = 4(1.05) = 4.20.$$

Thus,

$$P_0 = \frac{4.20}{0.11 - 0.05} = 70.$$

Hence,

$$\boxed{P_0 = \$70}.$$

5.3 Multi-Stage Dividend Model

Suppose dividends grow at g_1 for n years and subsequently at g_2 forever.

Then

$$P_0 = \sum_{t=1}^n \frac{D_0(1 + g_1)^t}{(1 + r_e)^t} + \frac{D_0(1 + g_1)^n(1 + g_2)}{(r_e - g_2)(1 + r_e)^n}.$$

5.4 H-Model

The H-model permits growth to decline linearly from g_S to g_L :

$$P_0 = \frac{D_0(1 + g_L)}{r_e - g_L} + \frac{D_0H(g_S - g_L)}{r_e - g_L}.$$

The H-model is useful when a firm is expected to transition gradually from abnormal growth to mature growth.

5.5 Three-Stage Dividend Model

A more flexible formulation is

$$P_0 = \sum_{t=1}^{T_1} \frac{D_t}{(1 + r_e)^t} + \sum_{t=T_1+1}^{T_2} \frac{D_t}{(1 + r_e)^t} + \frac{TV_{T_2}}{(1 + r_e)^{T_2}},$$

where

$$TV_{T_2} = \frac{D_{T_2+1}}{r_e - g_\infty}.$$

6 Free Cash Flow to Equity

FCFE measures cash available to common shareholders after operating requirements, investment, and net borrowing.

A commonly used expression is

$$FCFE = NI - (CapEx - D) - \Delta NWC.$$

Equivalently,

$$FCFE = NI - CapEx + Net \text{ Borrowing} - \Delta NWC.$$

Equity value is

$$P_0 = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1 + r_e)^t}.$$

6.1 Example

Suppose

$$NI = 100,$$

$$CapEx = 50,$$

$$D = 20,$$

and

$$\Delta NWC = 10.$$

Then

$$FCFE = 100 - (50 - 20) - 10 = 60.$$

If FCFE grows perpetually at 4% and

$$r_e = 10\%,$$

then

$$V_E = \frac{60(1.04)}{0.10 - 0.04} = 1040.$$

Thus,

$$\boxed{Equity\ Value = 1040}.$$

7 Free Cash Flow to Firm

FCFF represents cash available to all capital providers.

A common formulation is

$$FCFF = EBIT(1 - T) + D\&A - CapEx - \Delta NWC.$$

Enterprise value is

$$EV = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1 + WACC)^t}.$$

Then

$$Equity\ Value = EV - Debt + Cash.$$

7.1 Example

Suppose

$$EBIT = 150, \quad T = 25\%, \quad D\&A = 30,$$

$$CapEx = 40, \quad \Delta NWC = 15.$$

Then

$$FCFF = 150(0.75) + 30 - 40 - 15 = 87.5.$$

If

$$WACC = 8\%, \quad g = 3\%,$$

then

$$EV = \frac{87.5(1.03)}{0.08 - 0.03} = 1802.5.$$

If debt is 500 and cash is 200,

$$Equity\ Value = 1802.5 - 500 + 200 = 1502.5.$$

8 Economic Profit and EVA

Economic value added is

$$EVA = NOPAT - WACC \times Capital.$$

If invested capital is 800 and

$$NOPAT = 90, \quad WACC = 8\%,$$

then

$$EVA = 90 - 0.08(800) = 26.$$

A positive EVA means that accounting operating profit exceeds the capital charge. The enterprise value can be represented as

$$EV = \text{Invested Capital} + PV(\text{Future EVA}).$$

This formulation connects valuation to the economic-profit concept.

9 Residual Income Valuation

Residual income for equity is

$$RI_t = NI_t - r_e B_{t-1},$$

where B_{t-1} is beginning book equity.
Equity value is

$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1 + r_e)^t}$$

under the appropriate clean-surplus assumptions.

9.1 Example

Suppose

$$B_0 = 500,$$

and expected residual income is 30 annually forever, with

$$r_e = 10\%.$$

Then

$$V_0 = 500 + \frac{30}{0.10} = 800.$$

Thus,

$$\boxed{V_0 = 800}.$$

Residual-income valuation is particularly useful when dividends or free cash flow are difficult to interpret but accounting book values and earnings are informative.

10 Abnormal Earnings Growth

Another accounting-based approach values equity through expected earnings growth beyond the required return.

Conceptually,

$$V_0 = \text{Current Earnings Capitalization} + PV(\text{Abnormal Earnings Growth}).$$

This method is closely related to residual-income valuation but emphasizes earnings growth rather than the stock of book equity.

11 Price Multiples

Relative valuation asks:

$$\boxed{\text{What are comparable assets worth relative to a common economic metric?}}$$

11.1 Price-to-Earnings

$$P/E = \frac{Price}{EPS}.$$

If comparable companies trade at

$$P/E = 18$$

and a company has

$$EPS = 5,$$

then

$$P = 18(5) = 90.$$

11.2 Forward P/E

$$Forward\ P/E = \frac{P_0}{E[EPS_1]}.$$

This is preferable to trailing P/E when expected earnings are more economically relevant than historical earnings.

11.3 PEG Ratio

The PEG ratio is

$$PEG = \frac{P/E}{g}.$$

If

$$P/E = 20, \quad g = 10\%,$$

then, when growth is expressed as 10 rather than 0.10,

$$PEG = 2.$$

The ratio is heuristic rather than a complete valuation model.

11.4 Price-to-Book

$$P/B = \frac{Market\ Value\ of\ Equity}{Book\ Equity}.$$

For a bank with book value per share of 50 and a comparable P/B multiple of 1.5,

$$P = 1.5(50) = 75.$$

11.5 Price-to-Sales

$$P/S = \frac{Market\ Capitalization}{Sales}.$$

For sales of 1 billion and a comparable $P/S = 2$,

$$Market\ Capitalization = 2\ billion.$$

12 Enterprise Value Multiples

12.1 EV/EBITDA

$$EV/EBITDA = \frac{\text{Enterprise Value}}{EBITDA}.$$

If

$$EBITDA = 100$$

and the relevant multiple is 8,

$$EV = 800.$$

If debt is 250 and cash is 50,

$$\text{Equity Value} = 800 - 250 + 50 = 600.$$

12.2 EV/EBIT

$$EV/EBIT = \frac{EV}{EBIT}.$$

This can be preferable when depreciation represents economically meaningful consumption of capital.

12.3 EV/Sales

$$EV/Sales = \frac{EV}{Sales}.$$

This is frequently used for firms with low or negative earnings.

12.4 EV/FCFF

$$EV/FCFF = \frac{EV}{FCFF}.$$

This brings the multiple conceptually closer to discounted cash-flow valuation.

13 Comparable Company Analysis

The comparable-company method proceeds as follows:

1. identify economically comparable firms;
2. select the relevant valuation metric;
3. calculate peer multiples;
4. adjust for growth, profitability, leverage, and risk;
5. apply the relevant multiple;
6. convert enterprise value into equity value when necessary.

The fundamental relationship underlying a valuation multiple can be expressed schematically as

$$\text{Multiple} = f(g, r, ROIC, \text{risk}, \text{payout}, \text{capital intensity}).$$

Consequently, a multiple should not be treated as a universal constant.

14 Precedent Transactions

Precedent transactions value a company using multiples observed in completed acquisitions.

Typical metrics include

$$EV/EBITDA, \quad EV/EBIT, \quad EV/Sales.$$

Suppose recent acquisitions imply

$$EV/EBITDA = 10.$$

If the target has

$$EBITDA = 120,$$

then

$$EV = 1200.$$

If debt is 300 and cash is 100,

$$Equity Value = 1000.$$

The transaction multiple may contain a control premium and therefore cannot automatically be interpreted as a minority-market valuation.

15 Sum-of-the-Parts Valuation

Conglomerates can be valued by separately valuing their divisions:

$$EV_{SOTP} = \sum_{i=1}^n EV_i + Nonoperating Assets - Corporate Costs.$$

Suppose a company has three divisions:

$$EV_A = 500, \quad EV_B = 300, \quad EV_C = 200.$$

Then

$$EV_{SOTP} = 1000.$$

If net debt is 250,

$$Equity Value = 750.$$

SOTP can be especially useful when business segments have fundamentally different economics and therefore different appropriate valuation multiples.

16 Asset-Based Valuation

The asset-based approach values the company's assets and liabilities directly.

$$Equity Value = Fair Value of Assets - Fair Value of Liabilities.$$

It is particularly relevant for:

- financial institutions;
- natural-resource companies;
- investment companies;
- holding companies;
- asset-rich firms;
- liquidation situations.

16.1 Adjusted Net Asset Value

Suppose assets have fair values

$$A_1 = 300, \quad A_2 = 400, \quad A_3 = 200,$$

and liabilities are

$$L = 500.$$

Then

$$NAV = 300 + 400 + 200 - 500 = 400.$$

16.2 Liquidation Value

If assets cannot be sold at going-concern values, use liquidation discounts:

$$LV = \sum_i q_i A_i - L,$$

where q_i is the expected recovery fraction.

Suppose assets of 1,000 have an average recovery rate of 60% and liabilities are 400:

$$LV = 0.60(1000) - 400 = 200.$$

17 Replacement-Cost Valuation

Replacement-cost valuation asks:

What would it cost to recreate the firm's productive capacity?

A stylized expression is

$$V = \text{Replacement Cost} - \text{Economic Obsolescence} - \text{Physical Depreciation}.$$

This approach can be informative for infrastructure, utilities, industrial assets, and capital-intensive businesses.

18 Tobin's Q

Tobin's Q is conceptually

$$Q = \frac{\text{Market Value of Assets}}{\text{Replacement Cost of Assets}}.$$

If

$$Q > 1,$$

market valuation exceeds replacement cost.

If

$$Q < 1,$$

replacement cost exceeds market valuation.

The measure is associated with investment theory because firms may have incentives to expand productive capital when the market value of installed capital exceeds replacement cost.

19 Venture Capital Method

The venture-capital method begins with an expected exit value.

Suppose expected exit EBITDA is

$$EBITDA_T = 20$$

and the exit multiple is

$$10.$$

Then

$$Exit\ EV = 200.$$

If net debt at exit is zero,

$$Exit\ Equity = 200.$$

Suppose the investor requires a fivefold return and invests 20 today. The implied post-money equity value is

$$PostMoney = \frac{200}{5} = 40.$$

Thus the investor's initial ownership is

$$\frac{20}{40} = 50\%.$$

This approach embeds substantial assumptions about exit timing, exit multiples, dilution, and required return.

20 Probability-Weighted Valuation

Suppose a company has three possible outcomes:

Scenario	Probability	Equity Value
Failure	0.20	0
Moderate Success	0.50	100
Extreme Success	0.30	300

The probability-weighted value is

$$V = 0.20(0) + 0.50(100) + 0.30(300).$$

Therefore,

$$\boxed{V = 140}.$$

This is a basic application of expected-value theory.

21 Decision-Tree Valuation

A decision tree allows future decisions to respond to realized states.

For example,

$$V_0 = \frac{pV_{success} + (1-p)V_{failure}}{1+r}.$$

Unlike a simple scenario-weighted DCF, a decision tree can explicitly represent managerial decisions after uncertainty resolves.

This makes it useful for:

- drug development;
- mining;
- exploration;
- technology commercialization;
- restructuring;
- strategic investment.

22 Real Options Valuation

Equity valuation can contain embedded options.

Examples include:

- option to expand;
- option to abandon;
- option to delay;
- option to contract;
- option to switch technologies;
- option to enter a new market.

A simplified expansion option can be represented using the Black-Scholes structure:

$$C = SN(d_1) - Ke^{-rT}N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The variables have economic rather than merely financial interpretations:

Financial Option	Real-World Interpretation
Underlying asset	Present value of project opportunities
Strike price	Investment required
Volatility	Uncertainty about project value
Expiration	Period during which decision can be delayed
Risk-free rate	Time value of capital

23 Abandonment Option

A project with an abandonment option may be worth more than its static NPV.

Suppose

$$PV(Project) = 100$$

but liquidation value is

$$L = 60.$$

If the project can be abandoned when its continuation value falls below 60, the abandonment option creates a floor on recoverable value in certain states.

Thus,

$$V_{\text{project with option}} > V_{\text{static project}}$$

may hold.

The precise value requires state-contingent modeling.

24 Adjusted Present Value

The APV method separates operating value from financing effects:

$$APV = PV(\text{Unlevered Operations}) + PV(\text{Financing Side Effects})$$

For example,

$$APV = V_U + PV(\text{Tax Shields}) - PV(\text{Expected Financial Distress Costs}) + PV(\text{Other Financing Effects}).$$

This is particularly useful when leverage changes materially over time.

25 Leveraged Buyout Valuation

An LBO asks whether a business can support a transaction financed with substantial debt.

A simplified equity return calculation is

$$MOIC = \frac{\text{Exit Equity}}{\text{Initial Equity}}.$$

The internal rate of return satisfies

$$\text{Initial Equity} = \frac{\text{Exit Equity}}{(1 + IRR)^T}.$$

Thus,

$$IRR = \left(\frac{\text{Exit Equity}}{\text{Initial Equity}} \right)^{1/T} - 1.$$

25.1 Example

Suppose

$$\text{Initial Equity} = 200,$$

and after five years,

$$\text{Exit Equity} = 400.$$

Then

$$MOIC = 2.0.$$

The annualized return is

$$IRR = 2^{1/5} - 1 \approx 14.87\%.$$

26 Market Capitalization from Share Repurchases

Equity valuation must account for capital distributions and changes in share count.

If a company has

$$Net\ Income = 100$$

and repurchases shares worth 20, then the number of shares outstanding falls.

Consequently,

$$EPS = \frac{Net\ Income}{Shares}.$$

A repurchase can increase EPS without necessarily increasing intrinsic enterprise value.

This is an important distinction between:

per-share accounting metrics

and

economic value.

27 Implied Valuation and Reverse DCF

Instead of asking

What is the value given my assumptions?

reverse valuation asks

What assumptions are implied by today's market price?

Suppose a stock trades at 100.

One can solve for the growth rate g satisfying

$$100 = \frac{FCFF_1}{WACC - g}.$$

If

$$FCFF_1 = 8, \quad WACC = 10\%,$$

then

$$100 = \frac{8}{0.10 - g}.$$

Therefore,

$$0.10 - g = 0.08$$

and

$g = 2\%$.

Reverse DCF is therefore an expectation-extraction method rather than a conventional forecasting method.

28 Dividend Yield and Earnings Yield

Dividend yield is

$$DY = \frac{D_1}{P_0}.$$

Earnings yield is

$$EY = \frac{EPS}{P_0} = \frac{1}{P/E}.$$

For example, if

$$P = 100, \quad EPS = 8,$$

then

$$EY = 8\%.$$

These measures are useful for comparing equity claims with alternative assets, although they are not complete intrinsic valuation models.

29 Residual Asset and Liquidation Valuation

For distressed firms, going-concern valuation may be inappropriate.

A liquidation model can be written as

$$V_L = \sum_i Recovery_i - Administrative\ Costs - Debt - Other\ Claims.$$

The equity holder receives the residual:

$$Equity_L = \max(V_L, 0).$$

This naturally introduces an option-like structure because equity is a residual claim.

30 Equity as a Call Option on the Firm

Under the structural approach to credit risk, shareholders can be interpreted as holding a call option on the firm's assets.

Let

A = value of firm assets,

D = face value of debt.

At maturity,

$$E_T = \max(A_T - D, 0).$$

Therefore,

$$E_T = (A_T - D)^+$$

which is formally equivalent to a European call payoff.

This perspective links equity valuation to:

- option pricing;
- corporate finance;
- credit risk;
- capital structure;
- default theory.

31 Black–Scholes–Merton Structural Valuation

The equity value can be represented approximately as

$$E = AN(d_1) - De^{-rT}N(d_2),$$

where

$$d_1 = \frac{\ln(A/D) + (r + \sigma_A^2/2)T}{\sigma_A\sqrt{T}}$$

and

$$d_2 = d_1 - \sigma_A\sqrt{T}.$$

The model therefore connects:

asset value $\rightarrow$ equity value $\rightarrow$ default probability.

32 Conglomerate Discount and Holding-Company Valuation

A holding company may be valued as

$$V_H = \sum_i V_i - \text{Corporate Costs} - \text{Holding Discount}.$$

If

$$\sum_i V_i = 1000,$$

corporate costs have a present value of 50, and the market applies a 10% holding discount,

$$V_H = 1000(0.90) - 50 = 850.$$

Whether such a discount is economically justified depends on the underlying structure of ownership, taxation, governance, liquidity, capital allocation, and control.

33 Private Company Valuation

Private-company valuation combines several difficulties:

- limited market data;
- illiquidity;
- concentrated ownership;
- uncertain forecasts;
- key-person risk;
- unusual capital structures.

Possible approaches include:

$$DCF, \quad \text{Comparable Companies}', \quad \text{Precedent Transactions}', \quad \text{Asset Valuation}', \quad \text{Venture Capital}', \quad \text{Probability Weighted Valuation}.$$

Discounts or premiums for control and marketability must be handled carefully because they depend on the specific valuation premise.

34 Financial Institution Valuation

Banks and insurers require special treatment because debt-like instruments can be part of ordinary operations.

For banks, useful methods include

$$P/B, \quad P/E, \quad \text{Residual Income}.$$

A simplified bank residual-income model is

$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{ROE_t - r_e}{(1 + r_e)^t} B_{t-1}.$$

The key economic quantity is the spread

$$ROE - r_e.$$

A bank creating returns above its cost of equity generates positive economic value.

35 Insurance Company Valuation

Insurance valuation must consider:

- underwriting profits;
- investment income;
- reserve adequacy;
- capital requirements;
- catastrophe risk;
- policyholder behavior.

Embedded-option and scenario methods can become important because liabilities are themselves state-contingent.

36 Commodity and Resource Companies

For natural-resource firms,

$$NAV = PV(\text{Expected Resource Cash Flows})$$

can be combined with option values associated with exploration, development, expansion, and abandonment.

A mining company may therefore be decomposed as

$$V = PV(\text{Existing Mines}) + PV(\text{Development Projects}) + PV(\text{Exploration Options}) - Debt.$$

37 Technology Company Valuation

Technology firms frequently have:

- high intangible investment;
- uncertain growth;
- network effects;
- low current earnings;

- large option value.

A useful decomposition is

$$V = V_{\text{existing business}} + V_{\text{growth opportunities}}.$$

The second component can be represented using real-options or scenario-based methods.

38 Platform and Network Valuation

For platforms, value may depend on the number and interaction of users.

A stylized network-effect relationship is

$$V \propto N^\alpha,$$

where N is the number of participating users and $\alpha > 1$ can represent increasing network effects.

The economic challenge is determining whether the observed growth in users produces sustainable incremental cash flows.

39 Human Capital and Intangible Assets

Traditional accounting frequently fails to capitalize internally generated intangible investment fully.

An economic valuation may therefore reconstruct:

$$\frac{\text{Economic Capital}}{\text{Capital}} = \frac{\text{Physical Capital}}{\text{Capital}} + \frac{\text{Organizational Capital}}{\text{Capital}} + \frac{\text{Human Capital}}{\text{Capital}} + \frac{\text{Intellectual Property}}{\text{Property}}.$$

The corresponding cash-flow implications can be incorporated into FCFF, residual income, or asset-based models.

40 Growth Decomposition

Sustainable growth can be represented as

$$g = ROE \times Retention.$$

If

$$ROE = 15\%$$

and the retention ratio is

$$60\%,$$

then

$$g = 15\%(0.60) = 9\%.$$

This identity links valuation to accounting and corporate strategy.

A higher growth rate is not automatically valuable. Growth creates value when

$$ROIC > WACC.$$

If

$$ROIC < WACC,$$

growth can destroy economic value.

41 Value Creation and Competitive Advantage

A firm's intrinsic value can be decomposed conceptually into

$$V = V_{\text{assets in place}} + V_{\text{growth opportunities}}.$$

Growth opportunities are valuable when the firm can reinvest at

$$ROIC > WACC.$$

The economic-profit contribution of an incremental investment I is

$$\text{Economic Profit} = I(ROIC - WACC).$$

Thus,

$$\text{Growth creates value when incremental returns exceed the cost of capital.}$$

42 Terminal Value

Terminal value is often the largest component of a DCF.

The Gordon-growth formulation is

$$TV_T = \frac{FCF_{T+1}}{WACC - g}.$$

The exit-multiple formulation is

$$TV_T = \text{Metric}_T \times \text{Exit Multiple}.$$

A sound valuation should therefore test:

- terminal growth;
- WACC;
- exit multiple;
- reinvestment rate;
- terminal margins;
- terminal ROIC.

43 Sensitivity Analysis

Suppose

$$V = \frac{FCF_1}{WACC - g}.$$

The sensitivity of valuation to growth is

$$\frac{\partial V}{\partial g} = \frac{FCF_1}{(WACC - g)^2}.$$

The sensitivity to the discount rate is

$$\frac{\partial V}{\partial WACC} = -\frac{FCF_1}{(WACC - g)^2}.$$

Thus,

$$\frac{\partial V}{\partial g} = -\frac{\partial V}{\partial WACC}$$

in this simplified perpetual-growth model.

The closer g approaches $WACC$, the greater the valuation sensitivity.

44 Monte Carlo Valuation

Instead of assigning a single forecast to each variable, Monte Carlo valuation treats variables as random variables.

For example,

$$g \sim f_g, \quad WACC \sim f_r, \quad Margin \sim f_m.$$

For each simulation,

$$V^{(j)} = DCF(g^{(j)}, WACC^{(j)}, Margin^{(j)}, \dots).$$

After N simulations,

$$\hat{E}[V] = \frac{1}{N} \sum_{j=1}^N V^{(j)}.$$

One can also obtain quantiles:

$$P_5(V), \quad P_{50}(V), \quad P_{95}(V).$$

This converts valuation from a point estimate into a distribution.

45 Bayesian Valuation

A Bayesian approach begins with prior beliefs about uncertain parameters.

For an uncertain growth rate g ,

$$p(g \mid Data) \propto p(Data \mid g)p(g).$$

The posterior valuation is

$$E[V \mid Data] = \int V(g)p(g \mid Data) dg.$$

This provides a natural framework for incorporating:

- historical data;
- analyst forecasts;
- industry information;
- management guidance;
- macroeconomic information.

46 Factor-Based Relative Valuation

Instead of using one multiple, valuation can be modeled as a function of several characteristics:

$$P/E = \alpha + \beta_1 g + \beta_2 ROE - \beta_3 r + \beta_4 Payout + \varepsilon.$$

This converts comparable-company analysis into a cross-sectional statistical model.

A more general representation is

$$\ln(P_i) = \alpha + \beta^\top X_i + \varepsilon_i.$$

The fitted value can then provide an implied valuation for a target firm.

47 Regression-Based Valuation

A valuation regression may use

$$Y_i = \alpha + \beta_1 Growth_i + \beta_2 ROIC_i + \beta_3 Margin_i + \beta_4 Leverage_i + \varepsilon_i.$$

Potential dependent variables include

$$\ln(P/E), \quad \ln(EV/EBITDA), \quad \ln(P/B).$$

Important econometric problems include:

- heteroskedasticity;
- multicollinearity;
- endogeneity;
- omitted-variable bias;
- survivorship bias;
- look-ahead bias;
- selection bias.

48 Market-Implied Expected Return

If a stock price is observed and expected cash flows are estimated, one can solve for the discount rate.

For a Gordon model,

$$P_0 = \frac{D_1}{r - g}.$$

Therefore,

$$r = \frac{D_1}{P_0} + g.$$

Thus,

$\text{Required Return} = \text{Dividend Yield} + \text{Expected Growth}$

under the constant-growth assumptions.

49 Cross-Asset Relative Valuation

Equities can be compared with:

- government bonds;
- corporate bonds;
- real estate;
- commodities;
- private equity;
- infrastructure.

For example, the earnings yield is

$$EY = \frac{E}{P}.$$

A simplified equity-risk comparison is

$$ERP_{\text{implied}} \approx EY - r_f$$

although a rigorous implied-equity-risk-premium model requires explicit growth and cash-flow assumptions.

50 Behavioral Finance Perspective

Traditional valuation assumes that prices ultimately reflect fundamental information.

Behavioral finance adds possible deviations arising from:

- overconfidence;
- anchoring;
- representativeness;
- loss aversion;
- extrapolation;
- disposition effects;
- attention constraints.

A behavioral interpretation does not eliminate intrinsic valuation. Instead, it asks whether the market price may temporarily deviate from fundamental value.

51 Game-Theoretic Perspective

Firm value depends partly on the actions of competitors, regulators, suppliers, customers, and capital providers.

A strategic valuation can therefore be represented as

$$V_i = V_i(a_i, a_{-i}, s),$$

where

- a_i is the firm's strategy;
- a_{-i} represents competitors' strategies;
- s is the economic state.

Examples include:

- price wars;
- capacity expansion;
- technology standards;
- platform competition;
- merger strategies.

This links valuation to industrial organization and game theory.

52 Macroeconomic Perspective

Equity value can be expressed as a function of macroeconomic variables:

$$V = V(Y, \pi, r, u, FX, \dots),$$

where

Y = economic activity,

π = inflation,

r = interest rates,

u = unemployment,

and FX represents exchange rates.

The macroeconomic environment affects valuation through:

1. revenue growth;
2. operating margins;
3. financing costs;
4. taxes;
5. investment requirements;
6. terminal growth.

53 Accounting Perspective

Accounting provides the raw information from which many valuation variables are constructed:

$$Revenue \rightarrow EBITDA \rightarrow EBIT \rightarrow NOPAT \rightarrow FCFF.$$

Similarly,

$$Revenue \rightarrow Net\ Income \rightarrow EPS \rightarrow P/E.$$

And

$$Book\ Equity \rightarrow Residual\ Income \rightarrow Intrinsic\ Value.$$

The analyst must therefore distinguish carefully between:

Accounting profit

and

Economic profit.

54 Information-Theoretic Perspective

Valuation is fundamentally an information-compression problem.

A valuation model compresses a large information set

$$\mathcal{I} = \{\text{financial statements, prices, industry data, macro data, strategy, technology, regulation}\}$$

into a scalar

$$V = f(\mathcal{I}).$$

The quality of valuation therefore depends not merely on mathematical complexity but on whether the model preserves economically relevant information.

55 Uncertainty and Knightian Uncertainty

Risk can often be represented using a probability distribution:

$$X \sim F(x).$$

Knightian uncertainty arises when the probability distribution itself is uncertain.

One may therefore distinguish

Risk = known or estimable probabilities

from

Uncertainty = uncertainty about the probability model itself.

A practical valuation response is scenario analysis:

$$V = \sum_s p_s V_s$$

combined with model uncertainty:

$$p_s \in \mathcal{P}.$$

This produces a set of possible valuations rather than a single deterministic answer.

56 Real-Options and Strategic Management

Strategic investments often have asymmetric payoffs:

Downside $\approx$ Initial Investment,

while

Upside $\gg$ Initial Investment.

This asymmetry is one reason conventional NPV can understate the value of flexibility.

A strategic project may therefore be represented as

$$V = NPV_{\text{static}} + Option_{\text{expand}} + Option_{\text{delay}} + Option_{\text{abandon}}.$$

57 A Unified Valuation Framework

The major valuation families can be summarized as follows.

Method	Primary Object	Typical Application
DDM	Dividends	Mature dividend-paying firms
FCFE	Equity cash flow	Financially interpretable equity cash flows
FCFF	Enterprise cash flow	General operating companies
Residual Income	Accounting residual profit	Banks and firms with informative book values
EVA	Economic profit	Capital-intensive operating businesses
P/E	Earnings	Relative valuation
P/B	Book value	Banks and asset-intensive firms
P/S	Revenue	Early-stage or low-profit firms
EV/EBITDA	Operating cash-flow proxy	Corporate comparables
EV/EBIT	Operating profit	Asset-intensive firms
EV/Sales	Revenue	Low-profit/high-growth firms
Precedent Transactions	Acquisition prices	M&A valuation
SOTP	Segment values	Conglomerates
NAV	Asset values	Resources and holding companies
Liquidation Value	Recovery value	Distressed firms
Venture Capital	Exit value	Startups
LBO	Leveraged equity return	Buyouts
Real Options	Strategic flexibility	Projects with asymmetric opportunities
APV	Operating value plus financing effects	Changing leverage
Reverse DCF	Implied expectations	Market-price analysis
Monte Carlo	Distribution of values	High uncertainty
Bayesian Valuation	Posterior value	Uncertain parameters and information

58 Worked Integrated Example

Consider a hypothetical company, Alpha Corporation.
Suppose

$$Revenue = 1000,$$

$$EBITDA = 200,$$

$$EBIT = 150,$$

$$Tax = 25\%,$$

$$CapEx = 80,$$

$$D\&A = 50,$$

$$\Delta NWC = 20.$$

Then

$$NOPAT = 150(1 - 0.25) = 112.5.$$

FCFF is

$$FCFF = 112.5 + 50 - 80 - 20 = 62.5.$$

Suppose

$$WACC = 9\%, \quad g = 3\%.$$

Then

$$EV = \frac{62.5(1.03)}{0.09 - 0.03} = 1072.92.$$

If

$$Debt = 300, \quad Cash = 100,$$

then

$$Equity\ Value = 1072.92 - 300 + 100 = 872.92.$$

If diluted shares equal 20 million,

$$P_0 = \frac{872.92}{20} = 43.65.$$

Therefore,

$$\boxed{Intrinsic\ Value\ per\ Share \approx \$43.65}$$

under these assumptions.

58.1 Comparable Valuation

Suppose comparable companies trade at

$$EV/EBITDA = 7.$$

Then

$$EV = 7(200) = 1400.$$

Therefore,

$$Equity\ Value = 1400 - 300 + 100 = 1200.$$

Thus,

$$P = \frac{1200}{20} = 60.$$

The DCF and comparable valuation differ because they encode different assumptions and information.

58.2 Residual Income Valuation

Suppose

$$Book\ Equity = 600,$$

$$Net\ Income = 90,$$

and

$$r_e = 10\%.$$

Then residual income is

$$RI = 90 - 0.10(600) = 30.$$

If this residual income is expected to persist indefinitely,

$$V = 600 + \frac{30}{0.10} = 900.$$

Therefore,

$$P = \frac{900}{20} = 45.$$

We now have three different estimates:

$$DCF \approx \$43.65$$

$$Comparable\ valuation = \$60$$

$$Residual\ income = \$45$$

The appropriate analytical response is not to mechanically average them. Instead, the analyst should investigate why the models disagree.

59 Why Valuation Methods Disagree

Differences may arise from:

1. different growth assumptions;
2. different discount rates;
3. different terminal values;
4. different definitions of cash flow;
5. different capital structures;
6. accounting distortions;
7. peer-selection effects;
8. control premiums;
9. liquidity differences;
10. strategic-option values.

A useful diagnostic identity is

$$Valuation\ disagreement = Assumption\ disagreement + Measurement\ disagreement + Model\ disagreement.$$

60 Valuation Triangulation

A robust valuation process can use several independent methods:

$$V_1 = \text{DCF},$$

$$V_2 = \text{Relative Valuation},$$

$$V_3 = \text{Residual Income},$$

$$V_4 = \text{Asset Valuation},$$

$$V_5 = \text{Scenario/Option Valuation}.$$

The objective is not necessarily

$$V = \frac{1}{5} \sum_i V_i.$$

Instead, the analyst asks:

Which economic assumptions explain the dispersion among V_i ?

This transforms disagreement among valuation models into diagnostic information.

61 Common Valuation Errors

61.1 Terminal Growth Greater Than Discount Rate

The Gordon model requires

$$g < r.$$

Otherwise the denominator becomes non-positive.

61.2 Mixing FCFF and Cost of Equity

FCFF must generally be discounted using

$$WACC,$$

whereas FCFE should generally be discounted using

$$r_e.$$

61.3 Double Counting Cash

If enterprise value is calculated from operating cash flows, excess cash should generally be added separately when moving to equity value.

61.4 Ignoring Dilution

Options, restricted stock, convertible securities, and other claims can alter diluted shares.

61.5 Confusing Growth With Value Creation

Growth creates value only when incremental returns exceed the cost of capital:

$$ROIC > WACC.$$

61.6 Mechanical Use of Multiples

A multiple is not a fundamental law of valuation.

$$Multiple = f(\text{growth, risk, profitability, capital intensity, } \dots).$$

62 A Practical Valuation Workflow

A complete valuation process can be organized into twelve stages.

1. Understand the business model.
2. Reconstruct historical financial statements.
3. Normalize earnings and cash flows.
4. Estimate competitive economics.
5. Forecast revenue and margins.
6. Estimate reinvestment requirements.
7. Estimate the cost of capital.
8. Construct a DCF.
9. Conduct relative valuation.
10. Analyze residual income or economic profit.
11. Test strategic options and scenarios.
12. Perform sensitivity and uncertainty analysis.

The final output should contain both:

Estimated Value

and

Distribution of Plausible Values.

63 Interdisciplinary Synthesis

Equity valuation can be understood as an intersection of several disciplines.

Discipline	Contribution to Valuation
Finance	Discounting, risk, capital structure, asset pricing
Accounting	Earnings, book value, cash-flow reconstruction
Economics	Competition, growth, capital allocation, equilibrium
Econometrics	Forecasting, statistical inference, identification
Statistics	Probability distributions and uncertainty
Mathematics	Optimization, stochastic processes, option pricing
Game Theory	Strategic interaction and competitive behavior
Psychology	Investor biases and decision-making
Information Theory	Information compression and uncertainty
Computer Science	Simulation, machine learning, numerical methods
Strategy	Competitive advantage and strategic flexibility
Law	Ownership, contracts, regulation and claims priority
Political Economy	Taxation, institutions and regulatory risk
Operations Research	Resource allocation and decision trees

64 The Deep Structure of Equity Valuation

The different methods can ultimately be organized around four questions.

64.1 Question 1: What does the shareholder own?

The shareholder owns a residual claim on the economic resources of the enterprise:

$$Equity = Assets - Prior Claims.$$

64.2 Question 2: What economic benefits will the claim generate?

These can appear as

$$Dividends, \quad FCFE, \quad \text{Residual Income}, \quad \text{Liquidation Proceeds}, \quad \text{Strategic Option Payoffs}.$$

64.3 Question 3: When will those benefits occur?

Timing matters because

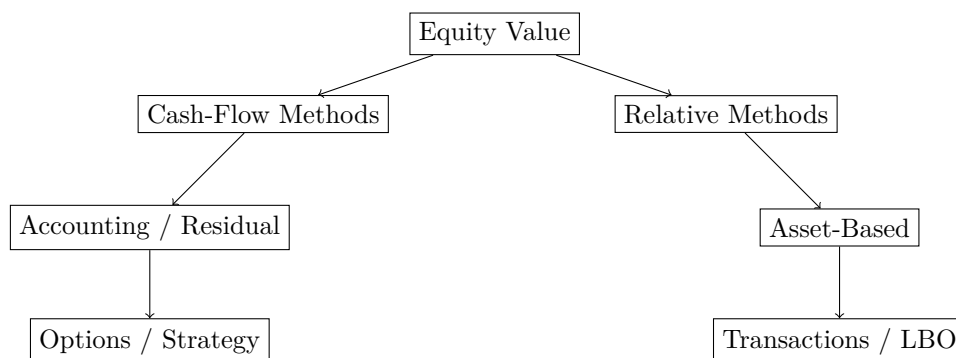
$$\$1 \text{ today} > PV(\$1 \text{ received later}).$$

64.4 Question 4: What risks and alternatives determine their value?

This incorporates

market risk, credit risk, business risk, strategic uncertainty, liquidity, regulation.

65 Master Valuation Map



The map emphasizes that equity valuation is not one formula. It is a family of models describing the same economic object from different perspectives.

66 Final Formula Sheet

Dividend Discount Model

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_e)^t}.$$

Gordon Growth

$$P_0 = \frac{D_1}{r_e - g}.$$

FCFE

$$V_E = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1+r_e)^t}.$$

FCFF

$$EV = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+WACC)^t}.$$

Equity Bridge

$$Equity = EV - Debt + Cash.$$

Residual Income

$$V_E = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r_e)^t}.$$

Economic Value Added

$$EVA = NOPAT - WACC \times Capital.$$

P/E

$$P = (P/E) \times EPS.$$

P/B

$$P = (P/B) \times BVPS.$$

EV/EBITDA

$$EV = (EV/EBITDA) \times EBITDA.$$

SOTP

$$EV_{SOTP} = \sum_i EV_i + Nonoperating\ Assets - Corporate\ Costs.$$

NAV

$$NAV = Fair\ Value\ Assets - Fair\ Value\ Liabilities.$$

APV

$$APV = PV(Unlevered\ Operations) + PV(Financing\ Effects).$$

LBO IRR

$$IRR = \left(\frac{Exit\ Equity}{Initial\ Equity} \right)^{1/T} - 1.$$

Real Option

$$C = SN(d_1) - Ke^{-rT}N(d_2).$$

Reverse DCF

Solve

$$P_{market} = DCF(\text{unknown assumption})$$

for the assumption implied by the observed market price.

67 Conclusion

The central lesson of equity valuation is that no valuation method should be understood as merely a mechanical formula.

Dividend models ask:

What are shareholders expected to receive?

DCF asks:

What cash-generating capacity does the enterprise possess?

Residual-income models ask:

How much profit is generated above the equity capital charge?

Relative valuation asks:

How is the market pricing economically comparable claims?

Asset valuation asks:

What are the underlying resources worth?

Transaction valuation asks:

What has strategic control over comparable assets been worth?

Option valuation asks:

What is the value of flexibility under uncertainty?

Reverse valuation asks:

What future does today's market price already assume?

The most comprehensive approach is therefore a triangulation:

$\text{Equity Valuation} = \text{Cash Flows} + \text{Accounting} + \text{Markets} + \text{Assets} + \text{Strategy} + \text{Uncertainty}$

subject to the consistency conditions imposed by finance, economics, accounting, statistics, and the institutional structure of the firm.

Ultimately, valuation is the disciplined transformation of uncertain future economic possibilities into present economic value.

The End